THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET C

Chapters 11 & 12: **Exploring Some Geometric Themes, Tales by Dots and Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In the Sierpinski triangle, the number of small shaded triangles after stage 4 is: 1

(a) 162 **(b)** 80 **(c)** 84 **(d)** 81

2. The number of edges of a cube is: 1

(a) 15 **(b)** 11 **(c)** 13 **(d)** 12

3. The number of edges of a triangular prism is: 1

(a) 9 **(b)** 8 **(c)** 12 **(d)** 10

4. The number of edges of a triangular pyramid (tetrahedron) is: 1

(a) 5 **(b)** 7 **(c)** 9 **(d)** 6

5. A solid has 5 faces and 6 vertices. Using $F + V - E = 2$, the number of edges is: 1

(a) 13 **(b)** 11 **(c)** 7 **(d)** 9

6. The mean of the data 8, 8, 10, 12, 14, 18 is: 1

(a) 12 **(b)** 11.6667 **(c)** 13.6667 **(d)** 9.66667

7. The mean of the data 45, 52, 38, 60, 55, 40 is: 1

(a) 46.3333 **(b)** 50.3333 **(c)** 49.5 **(d)** 48.3333

8. The median of 5, 6, 6, 9, 11, 13, 14 is: 1

(a) 12.1429 **(b)** 9 **(c)** 10 **(d)** 8

9. The median of 16, 18, 20, 22, 24, 54 is: 1

(a) 21 **(b)** 20 **(c)** 28.6667 **(d)** 22

10. The mean of 6 numbers is 12. 5 of them are 6, 9, 12, 15, 18. The remaining number is: 1

(a) 10 **(b)** 12 **(c)** 14 **(d)** 17

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. How many edges does a pentagonal pyramid have? 2

12. State two real-life examples of fractals found in nature. 2

13. Marks of 5 students: 40, 45, 50, 55, 60. Find the mean and median. 2

14. What are outliers? Give an example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. How many edges are there in a Koch curve after stage 3 if stage 0 is one segment? 3

16. Find the mean and median of the first ten natural numbers. 3

17. A cube of edge 3 cm is painted and cut into 27 unit cubes. How many cubes have exactly two faces painted? 3

OR

A cuboid has dimensions 8 cm × 5 cm × 3 cm. Find its total surface area. 3

18. The mean of five numbers is 12. If each number is increased by 3, find the new mean and new total. 3

OR

A bar graph and a line graph show the same data. State one difference in what they emphasise. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Find the number of small cubes in a $4 \times 4 \times 4$ cube painted outside. 5
(ii) How many have 3 faces painted?

OR

(i) A polyhedron has 6 faces and 12 edges. Find its vertices using Euler's formula. 5
(ii) Name two such solids.

20. (i) Describe how a line graph is read. 5
(ii) Name two real situations best shown by line graphs.

OR

(i) Show that the sum of deviations of values from their mean is zero for 4, 6, 8, 10, 12. 5
(ii) Explain the balancing idea.

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND